Nomenclature

Getting aligned on the words we're about to use

Quantum Resource Interface Workstream

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Quantum Resources

Matthias: + Status/Health – Resource is up, but calibration is needed.

A **Quantum Resource** is comprised of the QPU and the classical compute needed to execute requests coming in through an API (its endpoint)

openQSE(Quantum Computing Resource),
QDMI(Quantum device/backend),
QRMI(Quantum Resource)

Considerations for Working with Quantum Computers

- Scope — Can be a node-level resource or a shared, cluster-wide resource.
- Availability — Not always available; cannot accept jobs during calibration or other maintenance.
  - Calibration must not run while an HPC job is executing, since transpilation depends on calibration data — if it changes mid-job, transpilation would need to be re-run.
  - A mechanism is needed to block calibration while a job is running.
- Access – Mostly REST APIs
- Authentication — Accessing a quantum resource requires authentication.
- Cost — May be paid resources; usage must be tracked and capped to avoid runaway bills.
- Queueing — May have its own queue.

Scheduler/Resource Manager/Workload Manager

Workload Manager(WLM)

The name of the entire system that integrates the below two functions. e.g. Slurm etc.

Scheduler

DECIDE WHEN & WHERE

which pending job runs next, on which nodes, given priority, fairness, and backfill rules.

Resource Manager

TRACKS WHAT EXISTS

The inventory keeper: which nodes are up, how much CPU/memory/GPU each has, and what's currently in use.

Workflow, Hybrid-Workflow and Workflow Manager

Three core concepts in quantum-HPC integrated computing

Workflow

- A set of computational tasks automatically chained and executed based on their dependencies
- Expressed as a DAG (directed acyclic graph): pre-processing → computation → post-processing
- Emphasizes reproducibility, automation, and fault tolerance in execution

Hybrid-Workflow

- Combines classical (HPC) and quantum (QPU) resources within a single execution
- Classical side handles optimization and pre/post-processing; quantum side runs circuits
- Typical in iterative loop algorithms such as VQE and QAOA

Workflow Manager

- Software platform that manages task definitions, dependencies, and execution resources
- Automates scheduling, data hand-off, and re-execution after failures
- Abstracts and coordinates heterogeneous resources (CPU/GPU/QPU) under one interface

System Accounting Information

Doug Jacobsen: Time-series data(telemetry etc..) is not needed for scheduling. Usage is needed.

Accounting information refers to the records and data collected about resource usage, job history, and other relevant metrics of jobs executed on a HPC Cluster

Purpose of Accounting

- **Resource tracking:** Capture CPU, memory, GPU, I/O and other job-level resource usage.
- **Job history & auditability:** Record job submission, start time, end time, exit status, and execution metrics.
- **User/project visibility:** Track which user, group, and account submitted each job to understand utilization pattern.
- **Fairshare scheduling:** Provide historical usage data that feeds WLM's fairshare mechanism to set priority.
- **Reporting & analysis:** Allow system admin to generate usage reports across time intervals using tools.